

Amogh Samadhiya

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Summary

Software and Machine Learning Engineer specializing in distributed backend systems, MLOps orchestration, and scalable AI infrastructure. Experienced in architecting low-latency inference APIs, deploying containerized microservices, and automating data pipelines. Proficient in Python, C++17, and cloud-native observability. Seeking backend or machine learning engineering roles within highly technical engineering teams.

Education

Lakshmi Narain College of Technology (LNCT)

B.Tech in Electronics & Communication Engineering (CS Equivalent Track)

Bhopal, India

Expected May 2027

- **CGPA:** 7.67 / 10.0 | **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, System Design, ML.

Experience

Labmentix

Remote

Software & Machine Learning Engineering Intern

Jul 2025 - Jan 2026

- Architected a high-throughput inference API using FastAPI to process 10,000 daily machine learning requests, achieving strict sub-100ms P99 latency across distributed computing workloads.
- Containerized five core microservices with Docker and orchestrated multi-node Kubernetes deployments, ensuring high availability and zero-downtime rolling updates.
- Engineered robust CI/CD data pipelines via GitHub Actions to automate testing and deployment, compressing code release cycles from two hours to under 15 minutes.
- Integrated Prometheus observability and OpenTelemetry for distributed tracing to monitor real-time system health, enforcing backend reliability across 30 comprehensive peer code reviews.

Projects

Project Nexus (Marketplace Intelligence & RAG Platform) | *Flink, DuckDB, FAISS, LangChain* [GitHub](#) | [Live Demo](#)

- Engineered a declarative feature store utilizing Apache Flink and DuckDB, achieving 3.2ms P99 online feature lookup latency in Redis while guaranteeing point-in-time correct queries.
- Deployed a scalable RAG pipeline integrating LangChain and a FAISS vector database to retrieve and synthesize context asynchronously via FastAPI with a 67ms inference service-level agreement.
- Constructed an experimentation engine supporting variance reduction, orchestrated through Dagster to automate LLM evaluation metrics and model retraining triggers.

Voyage Analytics (MLOps Data Pipeline) | *FastAPI, MLflow, Airflow, K8s*

[GitHub](#) | [Live Demo](#)

- Built an automated MLflow evaluation framework to monitor model metrics across 15 Apache Airflow directed acyclic graphs, decreasing model update cycles by 60 percent.
- Designed resilient ETL data pipelines to ingest and validate distributed datasets, mitigating statistical data contamination during downstream model training.

Quantum Yield (Algorithmic Trading Platform) | *TensorFlow, VADER NLP, Docker*

[GitHub](#) | [Live Demo](#)

- Developed BiLSTM neural networks to forecast 8 global market indices, integrating VADER sentiment analysis, Monte Carlo value-at-risk simulations, and backtesting execution engines.
- Containerized the data ingestion and inference pipeline using Docker to stream real-time financial predictions directly to an interactive production dashboard.

EMIPredict AI (Financial Risk Assessment) | *XGBoost, Scikit-learn, Python*

[GitHub](#) | [Live Demo](#)

- Trained a dual-model classification architecture on 400,000 financial records to accurately assess loan risk profiles and calculate safe payment capacities.
- Established strict MLflow experiment tracking and validation constraints to monitor real-time feature decay, deploying the classification system securely as a web application.

Technical Skills

Languages & Core: Python, C++17, SQL, Bash, JavaScript, System Design, DSA, OOP, Database Management.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Large Language Models (LLMs), LangChain, RAG Pipelines, Vector Databases (FAISS), LLM Evaluation.

Data Engineering & MLOps: Apache Airflow, Apache Flink, Apache Spark, MLflow, Kafka, DuckDB, ETL Data Pipelines.
Backend & Cloud: FastAPI, Docker, Kubernetes, Redis, PostgreSQL, Prometheus Observability, Distributed Tracing (OpenTelemetry), GitHub Actions (CI/CD), AWS, gRPC.

Certifications

Oracle: GenAI Professional, AI Vector Search Professional. **AWS:** Cloud Architect, ML Foundations.